Gold Signal Desk — Full Website Code

Save this code as index.html, then drag the folder into Netlify Drop.

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<title>Gold Signal Desk</title>
<link href="https://fonts.googleapis.com/css2?family=Fraunces:opsz,wght@9..144,600&family=IBM+Plex+Sans:wght@400;500;600&display=swap" rel="s
tylesheet">
<style>
:root{--bg:#0e1923;--panel:#15232f;--line:#26394a;--ink:#e8edf1;--mute:#8da0b0;--brass:#d4a84a;--buy:#3fbf8f;--sell:#e2655a;--wait:#8da0b0;--
serif:'Fraunces',Georgia,serif;--sans:'IBM Plex Sans',system-ui,-apple-system,'Segoe UI',sans-serif}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:var(--ink);font:15px/1.5 var(--sans)}
header{display:flex;flex-wrap:wrap;gap:12px 24px;align-items:center;justify-content:space-between;padding:18px 24px;border-bottom:1px solid v
ar(--line)}
h1{font:600 22px var(--serif);margin:0;color:var(--brass)}
.controls{display:flex;flex-wrap:wrap;gap:10px;align-items:end}
label{display:flex;flex-direction:column;font-size:12px;color:var(--mute);gap:3px}
select,input,button{font:inherit;color:var(--ink);background:#0b141c;border:1px solid var(--line);border-radius:6px;padding:7px 10px}
input[type=range]{padding:0;width:130px}
button{cursor:pointer;background:var(--brass);color:#1b1405;border:0;font-weight:600}
button.ghost{background:transparent;color:var(--ink);border:1px solid var(--line)}
:focus-visible{outline:2px solid var(--brass);outline-offset:2px}
main{display:grid;gap:20px;padding:20px 24px;max-width:1280px;margin:auto}
@media(min-width:960px){main{grid-template-columns:380px 1fr}.wide{grid-column:1/-1}}
section{background:var(--panel);border:1px solid var(--line);border-radius:10px;padding:18px;min-width:0}
h2{font:600 16px var(--sans);margin:0 0 12px}
.verdict{font:600 64px/1 var(--serif);margin:4px 0 6px}
.buy{color:var(--buy)}.sell{color:var(--sell)}.wait{color:var(--wait)}
.price{color:var(--mute);font-variant-numeric:tabular-nums}
.why{color:var(--mute);margin:8px 0 0}
table{width:100%;border-collapse:collapse;font-variant-numeric:tabular-nums}
td,th{padding:7px 6px;border-bottom:1px solid var(--line);text-align:left;font-weight:400;vertical-align:top}
th{color:var(--mute);font-size:12px}
td.r,th.r{text-align:right}
.plan td:first-child{color:var(--mute)}
#chart{width:100%;height:380px;display:block}
.status{font-size:13px;color:var(--mute)}
.err{color:var(--sell)}
.stats{display:grid;grid-template-columns:repeat(auto-fit,minmax(130px,1fr));gap:12px}
.stat b{display:block;font:600 26px var(--serif)}
.stat span{color:var(--mute);font-size:12px}
.note{font-size:13px;color:var(--mute);max-width:78ch}
ul{margin:0;padding-left:18px;color:var(--mute)}
.tbl{overflow-x:auto}
.disclaimer{border-left:3px solid var(--brass);padding-left:12px;color:var(--mute);font-size:13px}
</style>
</head>
<body>
<header>
<h1>Gold Signal Desk</h1>
<div class="controls">
<label>Market<select id="sym"></select></label>
<label>Trading style<select id="mode">
<option value="scalp">Scalping (5m)</option>
<option value="day" selected>Day trading (15m)</option>
<option value="swing">Swing (4h)</option>
</select></label>
<label>Minimum score<input id="thr" type="range" min="40" max="85" value="60"></label>
<span id="thrv" class="status">60</span>
<label>Twelve Data key<input id="key" type="password" placeholder="optional: XAU/USD, forex"></label>
<button id="go">Analyze now</button>
</div>
</header>
<main>
<section aria-live="polite">
<div class="status" id="status">Loading…</div>
<div class="verdict wait" id="verdict">WAIT</div>
<div class="price" id="price"></div>
<p class="why" id="why"></p>
<h2 style="margin-top:18px">Trade plan</h2>
<table class="plan" id="plan"><tr><td>No setup right now.</td></tr></table>
<h2 style="margin-top:18px">Position size</h2>
<div class="controls">
<label>Account ($)<input id="acct" type="number" value="1000" min="1"></label>
<label>Risk per trade (%)<input id="risk" type="number" value="1" min="0.1" max="2" step="0.1"></label>
</div>
<p id="size" class="price"></p>
<p class="disclaimer">Educational tool only. Not financial advice. Trading can lose money. Test on a demo account first.</p>
</section>
<section>
<h2 id="chartTitle">Chart</h2>
<canvas id="chart" role="img" aria-label="Candlestick chart with EMAs and trade plan"></canvas>
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</section>
<section class="wide">
<h2>Why this signal</h2>
<div class="tbl"><table id="checks"></table></div>
</section>
<section class="wide">
<h2>Rule record on loaded candles</h2>
<div class="stats" id="bt"></div>
<p class="note" id="btnote"></p>
</section>
<section class="wide">
<h2>Rules</h2>
<ul>
<li>A signal needs trend, higher-timeframe agreement, and your minimum score.</li>
<li>Stop loss is beyond the recent swing, between 0.8 and 2.5 ATR.</li>
<li>Target 1 is 2R; target 2 is 3R.</li>
<li>Risk 1–2% per trade. Stop after your daily loss limit.</li>
<li>This app does not check economic news. Stay out during major releases.</li>
</ul>
</section>
</main>
<script>
const SYMS={
paxg:{label:'Gold (PAXG, free live data)',bin:'PAXGUSDT',td:'XAU/USD',p:4300},
xau:{label:'Gold XAU/USD (Twelve Data)',td:'XAU/USD',p:4300},
eur:{label:'EUR/USD (Twelve Data)',td:'EUR/USD',p:1.08},
btc:{label:'Bitcoin (free live data)',bin:'BTCUSDT',td:'BTC/USD',p:90000},
eth:{label:'Ethereum (free live data)',bin:'ETHUSDT',td:'ETH/USD',p:3000}
};
const MODES={scalp:{tf:'5m',htf:'1h',label:'Scalping'},day:{tf:'15m',htf:'4h',label:'Day trading'},swing:{tf:'4h',htf:'1d',label:'Swing'}};
const TD={'5m':'5min','15m':'15min','1h':'1h','4h':'4h','1d':'1day'};
const MS={'5m':3e5,'15m':9e5,'1h':36e5,'4h':144e5,'1d':864e5};
const BIN_HOSTS=['https://data-api.binance.vision','https://api.binance.com'];
const REFRESH_MS=60000, COST=0.0002;
const $=s=>document.querySelector(s);
let state=null,timer=null,isDemo=false;
$('#sym').innerHTML=Object.entries(SYMS).map(([k,v])=>`<option value="${k}">${v.label}</option>`).join('');
$('#key').value=localStorage.getItem('td_key')||'';
$('#key').oninput=e=>localStorage.setItem('td_key',e.target.value.trim());

async function fetchCandles(sym,iv,n){
const s=SYMS[sym],key=$('#key').value.trim();
if(s.bin){
let lastErr;
for(const host of BIN_HOSTS){
try{
const r=await fetch(`${host}/api/v3/klines?symbol=${s.bin}&interval=${iv}&limit=${n}`);
if(!r.ok)throw new Error('HTTP '+r.status);
const j=await r.json();
return j.map(k=>({t:k[0],o:+k[1],h:+k[2],l:+k[3],c:+k[4]}));
}catch(e){lastErr=e}
}
throw lastErr;
}
if(!key)throw new Error('This market needs a free Twelve Data API key.');
const r=await fetch(`https://api.twelvedata.com/time_series?symbol=${encodeURIComponent(s.td)}&interval=${TD[iv]}&outputsize=${n}&apikey=${ke
y}`);
const j=await r.json();
if(j.status==='error'||!j.values)throw new Error(j.message||'No data returned');
return j.values.reverse().map(v=>({t:Date.parse(v.datetime.replace(' ','T')+(v.datetime.length>10?'Z':'T00:00:00Z')),o:+v.open,h:+v.high,l:+v
.low,c:+v.close}));
}

function demoData(n,iv,sym){
let p=SYMS[sym].p,t=Date.now()-n*MS[iv];const out=[];
for(let k=0;k<n;k++){
const o=p,drift=Math.sin(k/45)*0.0005,c=o*(1+drift+(Math.random()-.5)*0.003);
out.push({t,o,h:Math.max(o,c)*(1+Math.random()*0.0012),l:Math.min(o,c)*(1-Math.random()*0.0012),c});
p=c;t+=MS[iv];
}
return out;
}

function ema(a,p){const k=2/(p+1),r=[];let e=a[0];for(let i=0;i<a.length;i++){e=i?a[i]*k+e*(1-k):a[0];r.push(e)}return r}
function rsi(c,p=14){
const r=Array(c.length).fill(50);let g=0,l=0;
for(let i=1;i<c.length;i++){
const d=c[i]-c[i-1],u=Math.max(d,0),w=Math.max(-d,0);
if(i<=p){g+=u/p;l+=w/p}else{g=(g*(p-1)+u)/p;l=(l*(p-1)+w)/p}
if(i>=p)r[i]=l===0?100:100-100/(1+g/l);
}
return r;
}
function atr(d,p=14){
const r=[];let a=0;
for(let i=0;i<d.length;i++){
const tr=i?Math.max(d[i].h-d[i].l,Math.abs(d[i].h-d[i-1].c),Math.abs(d[i].l-d[i-1].c)):d[i].h-d[i].l;
a=i<p?(a*i+tr)/(i+1):(a*(p-1)+tr)/p;r.push(a);
}
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return r;
}
function calc(d){
const c=d.map(x=>x.c),e12=ema(c,12),e26=ema(c,26),m=e12.map((v,i)=>v-e26[i]),sg=ema(m,9);
return{e20:ema(c,20),e50:ema(c,50),e200:ema(c,200),mh:m.map((v,i)=>v-sg[i]),rs:rsi(c),at:atr(d)};
}

function score(d,I,i){
const x=d[i],c=x.c,a=I.at[i];let bu=0,be=0;const comps=[];
const add=(n,u,w,t)=>{comps.push({n,u,w,t});bu+=u;be+=w};
const up=c>I.e50[i]&&I.e50[i]>I.e200[i],dn=c<I.e50[i]&&I.e50[i]<I.e200[i];
add('Trend 50/200 EMA',up?30:c>I.e200[i]?10:0,dn?30:c<I.e200[i]?10:0,up?'Uptrend':dn?'Downtrend':'Mixed');
const w5=d.slice(i-4,i+1),lo5=Math.min(...w5.map(z=>z.l)),hi5=Math.max(...w5.map(z=>z.h));
const nearL=Math.min(Math.abs(lo5-I.e20[i]),Math.abs(lo5-I.e50[i]))<=0.6*a;
const nearH=Math.min(Math.abs(hi5-I.e20[i]),Math.abs(hi5-I.e50[i]))<=0.6*a;
const pb=up&&nearL&&c>I.e20[i],ps=dn&&nearH&&c<I.e20[i];
add('Pullback to EMA 20/50',pb?20:0,ps?20:0,pb?'Dip bought at EMA':ps?'Rally sold at EMA':'No pullback');
const mu=I.mh[i]>0&&I.mh[i]>I.mh[i-1],md=I.mh[i]<0&&I.mh[i]<I.mh[i-1],r=I.rs[i];
add('Momentum MACD + RSI',(mu?15:0)-(r>75?10:0),(md?15:0)-(r<25?10:0),`MACD ${mu?'rising':md?'falling':'flat'}, RSI ${r.toFixed(0)}`);
let sw=0,sb=0;
for(let k=0;k<6;k++){
const j=i-k,ref=d.slice(j-20,j);
const pl=Math.min(...ref.map(z=>z.l)),ph=Math.max(...ref.map(z=>z.h));
if(d[j].l<pl&&d[j].c>pl)sw=20;
if(d[j].h>ph&&d[j].c<ph)sb=20;
}
add('Liquidity sweep',sw,sb,sw?'Lows swept, reclaimed':sb?'Highs swept, rejected':'None');
const h20=Math.max(...d.slice(i-20,i).map(z=>z.h)),l20=Math.min(...d.slice(i-20,i).map(z=>z.l));
add('Break of structure',c>h20?10:0,c<l20?10:0,c>h20?'Broke 20-bar high':c<l20?'Broke 20-bar low':'Inside range');
let fu=0,fd=0;
for(let k=i-9;k<=i;k++){if(d[k].l>d[k-2].h)fu=5;if(d[k].h<d[k-2].l)fd=5}
add('Fair value gap',fu,fd,fu?'Bullish gap':fd?'Bearish gap':'None');
const hr=new Date(x.t).getUTCHours(),ses=hr>=7&&hr<20?5:0;
add('London / New York session',ses,ses,ses?'Active session':'Quiet session');
const dir=bu>=be?1:-1,best=Math.max(bu,be),worst=Math.min(bu,be),conf=best-0.5*worst;
const aligned=dir===1?up:dn;
const wk=d.slice(i-7,i+1);
let risk=dir===1?c-(Math.min(...wk.map(z=>z.l))-0.2*a):(Math.max(...wk.map(z=>z.h))+0.2*a)-c;
risk=Math.min(Math.max(risk,0.8*a),2.5*a);
return{dir,conf,aligned,comps,entry:c,risk,atr:a,sl:c-dir*risk,tp1:c+dir*2*risk,tp2:c+dir*3*risk};
}

function backtest(d,I,thr){
let n=0,w=0,R=0,busyUntil=-1;
for(let i=210;i<d.length-2;i++){
if(i<busyUntil)continue;
const s=score(d,I,i);
if(!(s.conf>=thr&&s.aligned))continue;
const spread=s.entry*COST;
const sl=s.dir===1?s.sl-spread:s.sl+spread;
const tp1=s.dir===1?s.tp1-spread:s.tp1+spread;
let res=null,j=i+1;
for(;j<Math.min(i+49,d.length);j++){
const z=d[j];
if(s.dir===1){if(z.l<=sl){res=-1;break}if(z.h>=tp1){res=2;break}}
else{if(z.h>=sl){res=-1;break}if(z.l<=tp1){res=2;break}}
}
if(res===null){j=Math.min(j,d.length-1);res=((d[j].c-s.entry)*s.dir/s.risk)-1}
n++;if(res>0)w++;R+=res;busyUntil=j+1;
}
return{n,w,R};
}

async function run(demo){
isDemo=!!demo;
const sym=$('#sym').value,m=MODES[$('#mode').value],thr=+$('#thr').value;
$('#status').textContent='Loading market data…';
let d,h;
try{
if(demo){d=demoData(500,m.tf,sym);h=demoData(300,m.htf,sym)}
else[d,h]=await Promise.all([fetchCandles(sym,m.tf,500),fetchCandles(sym,m.htf,300)]);
}catch(e){
$('#status').textContent='Could not load live data: '+e.message;
const b=document.createElement('button');b.className='ghost';b.textContent='Show demo data';b.onclick=()=>run(true);$('#status').append(b);
return;
}
const live=d[d.length-1],cd=d.slice(0,-1),hc=h.slice(0,-1);
const I=calc(cd),H=calc(hc),i=cd.length-1,hi=hc.length-1;
const s=score(cd,I,i);
const hdir=H.e50[hi]>H.e200[hi]?1:H.e50[hi]<H.e200[hi]?-1:0;
const htfOk=hdir!==0&&hdir===s.dir;
const go=s.conf>=thr&&s.aligned&&htfOk;
const bt=backtest(cd,I,thr);
state={d,I,s,go,m,sym,thr,bt,htfDir:hdir,live};
render();
$('#status').textContent=(isDemo?'DEMO DATA. ':'Live. ')+'Updated '+new Date().toLocaleTimeString()+'. Closed candles only.';
clearTimeout(timer);if(!isDemo)timer=setTimeout(()=>run(false),REFRESH_MS);
}
```

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const fmt=v=>v>=100?v.toFixed(2):v.toFixed(5);
function render(){
const{d,I,s,go,m,sym,thr,bt,htfDir,live}=state;
const v=$('#verdict');
v.textContent=go?(s.dir===1?'BUY':'SELL'):'WAIT';
v.className='verdict '+(go?(s.dir===1?'buy':'sell'):'wait');
$('#price').textContent=`${SYMS[sym].label.split(' (')[0]} ${fmt(live.c)} · ${m.label} on ${m.tf}`;
const why=[];
if(s.conf<thr)why.push(`Score ${s.conf.toFixed(0)} is below ${thr}.`);
if(!s.aligned)why.push('Price is not in a clean trend for this direction.');
if(htfDir===0)why.push(`The ${m.htf} trend is neutral.`);
else if(htfDir!==s.dir)why.push(`The ${m.htf} trend disagrees.`);
$('#why').textContent=go?`Score ${s.conf.toFixed(0)} of 105; entry timeframe and ${m.htf} agree.`:why.join(' ')+' No trade is the correct tra
de.';
const rows=go?[['Entry (market)',s.entry],['Stop loss',s.sl],['Target 1 (2R)',s.tp1],['Target 2 (3R)',s.tp2],['Risk per unit',s.risk]]:[];
$('#plan').innerHTML=rows.length?rows.map(r=>`<tr><td>${r[0]}</td><td class="r">${fmt(r[1])}</td></tr>`).join(''):'<tr><td>No setup right now
.</td></tr>';
sizing();
$('#checks').innerHTML='<tr><th>Strategy</th><th>Chart reading</th><th class="r">Points</th></tr>'+s.comps.map(c=>{
const p=s.dir===1?c.u:c.w;return`<tr><td>${c.n}</td><td>${c.t}</td><td class="r">${p}</td></tr>`}).join('')+
`<tr><td><b>Total</b></td><td></td><td class="r"><b>${s.conf.toFixed(0)}</b></td></tr>`;
const wr=bt.n?bt.w/bt.n*100:0;
$('#bt').innerHTML=[['Signals',bt.n],['Win rate',bt.n?wr.toFixed(0)+'%':'-'],['Total result',(bt.R>=0?'+':'')+bt.R.toFixed(1)+'R'],['Average
per trade',bt.n?(bt.R/bt.n).toFixed(2)+'R':'-']].map(x=>`<div class="stat"><b>${x[1]}</b><span>${x[0]}</span></div>`).join('');
$('#btnote').textContent=`Same rules replayed on the last ${d.length-1} candles of ${m.tf}, including a 0.02% cost estimate. Past results do
not guarantee future results.`;
$('#chartTitle').textContent=`Chart · ${m.tf}`;
draw();
}

function sizing(){
if(!state)return;
const{s,go,sym}=state,acct=+$('#acct').value,rp=+$('#risk').value;
if(!go){$('#size').textContent='Appears when a signal is active.';return}
const units=acct*rp/100/s.risk;
$('#size').textContent=`Risking $${(acct*rp/100).toFixed(2)} → ${units.toFixed(units<1?4:2)} units`+(SYMS[sym].td==='XAU/USD'?` (${(units/100
).toFixed(3)} lots of 100 oz)`:'');
}

function draw(){
if(!state)return;
const{d,I,s,go}=state,cv=$('#chart'),dpr=devicePixelRatio||1,W=cv.clientWidth,H=cv.clientHeight;
cv.width=W*dpr;cv.height=H*dpr;const g=cv.getContext('2d');g.scale(dpr,dpr);
const N=Math.min(90,d.length),off=d.length-N,v=d.slice(off);
let hi=Math.max(...v.map(z=>z.h)),lo=Math.min(...v.map(z=>z.l));
if(go){hi=Math.max(hi,s.tp2,s.sl);lo=Math.min(lo,s.tp2,s.sl)}
const pad=(hi-lo)*.05;hi+=pad;lo-=pad;
const y=p=>H-16-(p-lo)/(hi-lo)*(H-32),cw=(W-70)/N,x=k=>6+k*cw+cw/2;
const css=getComputedStyle(document.documentElement),col=n=>css.getPropertyValue(n).trim();
g.font='11px '+col('--sans');g.lineWidth=1;
v.forEach((z,k)=>{
const upc=z.c>=z.o;g.strokeStyle=g.fillStyle=upc?col('--buy'):col('--sell');
g.beginPath();g.moveTo(x(k),y(z.h));g.lineTo(x(k),y(z.l));g.stroke();
g.fillRect(x(k)-cw*.35,Math.min(y(z.o),y(z.c)),cw*.7,Math.max(1,Math.abs(y(z.o)-y(z.c))));
});
[['e20','#8da0b0'],['e50',col('--brass')],['e200','#6c8fd0']].forEach(([key,cl])=>{
g.strokeStyle=cl;g.beginPath();
v.forEach((_,k)=>{const q=I[key][off+k];k?g.lineTo(x(k),y(q)):g.moveTo(x(k),y(q))});g.stroke();
});
if(go){
[['Entry',s.entry,'#e8edf1'],['Stop',s.sl,col('--sell')],['T1',s.tp1,col('--buy')],['T2',s.tp2,col('--buy')]].forEach(([n,p,cl])=>{
g.strokeStyle=g.fillStyle=cl;g.setLineDash([5,4]);g.beginPath();g.moveTo(0,y(p));g.lineTo(W-64,y(p));g.stroke();g.setLineDash([]);
g.fillText(n+' '+fmt(p),W-62,y(p)+4);
});
}
}
addEventListener('resize',draw);
$('#go').onclick=()=>run(false);
['sym','mode'].forEach(id=>$('#'+id).onchange=()=>run(false));
$('#thr').oninput=e=>{$('#thrv').textContent=e.target.value};
$('#thr').onchange=()=>run(false);
['acct','risk'].forEach(id=>$('#'+id).oninput=sizing);
run(false);
</script>
</body>
</html>
```